

Practice Exercises

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1 Scheduling Jobs with Cron & Anacron

- List all scheduled cron jobs for the current user.
- Schedule a cron job to run a script /home/user/backup.sh every day at 2 AM.
- Create an anacron job that runs a script /home/user/cleanup.sh every 3 days.
- View the system-wide crontab file. Where are additional cron jobs stored?

2 Archiving & Transferring Files

- Create a tar archive of /home/user/documents and name it backup.tar.
- Compress backup.tar using gzip. What is the final filename?
- Extract the contents of backup.tar.gz into /home/user/recovered.

3 Understanding the Boot Process

- Identify the current system boot time using who.
- View the GRUB configuration file location.
- List the last 5 boot logs from the system journal.

4 Recovering Root Password

- What key should be pressed to interrupt the bootloader during startup?
- Which command allows modifying the kernel parameters to reset the root password?
- What must be done after resetting the root password to ensure proper relabeling?

Week 11 practical/lab answers

These directly answer the practice exercises on the final slide.

Practical 1 – Cron and Anacron

Question 1: List all scheduled Cron jobs for the current user.

```
crontab -l
```

Possible result when no jobs exist:

```
no crontab for user
```

Question 2: Run `/home/user/backup.sh` every day at 2:00 AM.

Open the current user's crontab:

```
crontab -e
```

Add:

```
0 2 * * * /home/user/backup.sh
```

Save and exit.

Verify:

```
crontab -l
```

The script must be executable:

```
chmod +x /home/user/backup.sh
```

Question 3: Create an Anacron job that runs `/home/user/cleanup.sh` every three days.

Edit:

```
sudo vi /etc/anacrontab
```

Add:

```
3 10 cleanup-job /home/user/cleanup.sh
```

Meaning:

- `3` — run every three days.
- `10` — wait ten minutes after Anacron starts.
- `cleanup-job` — unique identifier.
- Final field — script to execute.

Make the script executable:

```
chmod +x /home/user/cleanup.sh
```

The slide specifies the three-day period but not the delay value, so `10` minutes is a reasonable practical value rather than a uniquely required answer.

Question 4: View the system-wide crontab. Where are additional jobs stored?

```
cat /etc/crontab
```

Additional system cron jobs are stored in:

```
/etc/cron.d
```

Other periodic script directories include:

```
/etc/cron.hourly
```

```
/etc/cron.daily
```

/etc/cron.weekly

Practical 2 – Archiving files

Question 1: Create `backup.tar` from `/home/user/documents`.

```
tar -cf backup.tar /home/user/documents
```

Verbose version:

```
tar -cvf backup.tar /home/user/documents
```

Verify:

```
ls -lh backup.tar  
tar -tf backup.tar
```

A message about removing the leading `/` is normal when archiving an absolute path. It prevents extraction from automatically writing directly into the root filesystem.

Question 2: Compress `backup.tar` using `gzip`. What is the final filename?

```
gzip backup.tar
```

Final filename:

```
backup.tar.gz
```

Verify:

```
ls -lh backup.tar.gz
```

Question 3: Extract `backup.tar.gz` into `/home/user/recovered`.

Create the destination:

```
mkdir -p /home/user/recovered
```

Extract:

```
tar -xzf backup.tar.gz -C /home/user/recovered
```

Verify:

```
find /home/user/recovered
```

Because the archive was created using the path `/home/user/documents`, the extracted structure may contain:

```
/home/user/recovered/home/user/documents
```

A cleaner archive can be created using:

```
tar -cf backup.tar -C /home/user documents
```

Then extraction places `documents` directly under the destination.

Practical 3 – Boot process

Question 1: Identify the current system boot time using `who`.

```
who -b
```

Question 2: View the GRUB configuration file location.

Main generated configuration:

```
/boot/grub2/grub.cfg
```

View it:

```
sudo less /boot/grub2/grub.cfg
```

Administrator defaults are stored in:

```
/etc/default/grub
```

Question 3: List the last five boot logs from the journal.

```
journalctl --list-boots | tail -n 5
```

To view one of those boots:

```
journalctl -b -1
```

Here, `-1` means the previous boot.

Practical 4 – Recovering the root password

This procedure should only be performed on a Linux machine you own or are authorised to administer.

Question 1: What key interrupts the bootloader during startup?

Press **any key except Enter** during the GRUB countdown.

The slides then instruct you to highlight the desired entry and press:

e

to edit it.

Question 2: Which command or parameter allows the kernel settings to be modified for password recovery?

At the GRUB menu:

1. Press **e** to edit the selected entry.
2. Locate the line beginning with **linux**.
3. Append:

rd.break

4. Press:

Ctrl+X

to boot with the modified parameters.

Complete recovery sequence

At the emergency shell:

```
mount -o remount,rw /sysroot
```

Enter the installed system:

```
chroot /sysroot
```

Reset the root password:

```
passwd root
```

Enter the new password twice.

Request SELinux relabelling:

```
touch /.autorelabel
```

Exit twice:

```
exit
```

```
exit
```

The system then continues booting or restarts, and SELinux relabels the affected files.

Question 3: What must be done after resetting the root password?

Create:

```
touch /.autorelabel
```

This ensures that files changed during password recovery, including `/etc/shadow`, receive the correct SELinux labels during the next boot.

Complete Week 11 lab workflow

Part A – Create test scripts

```
mkdir -p ~/week11-lab
```

```
cd ~/week11-lab
```

Create a backup script:

```
vi backup.sh
```

Content:

```
#!/bin/bash
```

```
date >> /home/user/backup-log.txt
```

Make it executable:

```
chmod +x backup.sh
```

Test it:

```
./backup.sh
```

```
cat /home/user/backup-log.txt
```

Part B – Schedule it with Cron

```
crontab -e
```

Add:

```
0 2 * * * /home/user/week11-lab/backup.sh
```

Verify:

```
crontab -l
```

```
systemctl status crond
```

For immediate testing, temporarily schedule it every minute:

```
* * * * * /home/user/week11-lab/backup.sh
```

Wait slightly over one minute, then check:

```
cat /home/user/backup-log.txt
```

Return it to the required 2:00 AM schedule after testing.

Part C – Archive and recover files

```
mkdir -p ~/week11-lab/documents
```

```
echo "Report one" > ~/week11-lab/documents/report1.txt
```

```
echo "Report two" > ~/week11-lab/documents/report2.txt
```

Create archive:

```
tar -cf ~/week11-lab/backup.tar -C ~/week11-lab documents
```

List contents:

```
tar -tf ~/week11-lab/backup.tar
```

Compress:

```
gzip ~/week11-lab/backup.tar
```

Extract:

```
mkdir -p ~/week11-lab/recovered  
tar -xzf ~/week11-lab/backup.tar.gz \  
-C ~/week11-lab/recovered
```

Verify:

```
find ~/week11-lab/recovered  
cat ~/week11-lab/recovered/documents/report1.txt
```

Part D – Inspect boot information

```
who -b  
journalctl --list-boots | tail -n 5  
cat /etc/default/grub  
sudo less /boot/grub2/grub.cfg
```

Press **q** to exit **less**.

Common Week 11 mistakes

Putting the fields in the wrong Cron order

Correct order:

minute hour day month weekday command

For 2:00 AM:

0 2 * * * command

Not:

2 0 * * * command

The latter means 12:02 AM.

Including a username in a user crontab

Incorrect inside `crontab -e`:

```
0 2 * * * user /home/user/backup.sh
```

Correct:

```
0 2 * * * /home/user/backup.sh
```

The username field is only included in system crontab files.

Using a script that is not executable

Fix:

```
chmod +x /home/user/backup.sh
```

Also include a shebang:

```
#!/bin/bash
```

Using relative paths in Cron

Avoid:

```
0 2 * * * backup.sh
```

Use:

```
0 2 * * * /home/user/backup.sh
```

Cron has a limited environment.

Confusing archiving and compression

```
tar -cf backup.tar documents
```

creates an archive.

```
gzip backup.tar
```

compresses it into:

```
backup.tar.gz
```

Forgetting to create the extraction directory

Before:

```
tar -xzf backup.tar.gz -C /home/user/recovered
```

run:

```
mkdir -p /home/user/recovered
```

Editing **grub.cfg** directly

The slides state that:

```
/boot/grub2/grub.cfg
```

should not normally be manually edited.

Modify:

```
/etc/default/grub
```

or scripts in:

```
/etc/grub.d
```

then regenerate the configuration.

Forgetting SELinux relabelling after password recovery

Run:

```
touch /.autorelabel
```

before exiting the chroot.

Week 11 essential commands

Command	Purpose
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<code>crontab -l</code>	List current user's Cron jobs
<code>crontab -e</code>	Edit current user's Cron jobs
<code>crontab -r</code>	Remove current user's crontab
<code>systemctl status crond</code>	Check the Cron daemon
<code>cat /etc/crontab</code>	View system-wide Cron configuration
<code>tar -cf</code>	Create an archive
<code>tar -tf</code>	List archive contents
<code>tar -xf</code>	Extract an archive
<code>gzip</code>	Compress using gzip
<code>gunzip</code>	Decompress gzip
<code>tar -czf</code>	Create gzip-compressed archive
<code>tar -xzf</code>	Extract gzip-compressed archive
<code>who -b</code>	Display system boot time
<code>journalctl --list-boots</code>	List recorded boot sessions
<code>journalctl -b</code>	View a boot journal
<code>cat /etc/default/grub</code>	View GRUB administrator settings
<code>grub2-mkconfig</code>	Regenerate the GRUB configuration
<code>rd.break</code>	Interrupt boot before switching to the installed root
<code>chroot /sysroot</code>	Enter the installed filesystem during recovery
<code>touch /.autorelabel</code>	Request SELinux relabelling